

Landslide Safety Guide

Do's and Don'ts — Before, During, and After a Landslide

1. Understanding Landslides

A landslide is the downhill movement of rock, soil, and debris under the pull of gravity. Movement can happen gradually as a slow creep, or suddenly and without warning as a fast-moving debris flow capable of destroying homes and roads in seconds. Landslides are often triggered by, or occur alongside, other hazards — heavy or prolonged rainfall, earthquakes, rapid snowmelt, volcanic activity, wildfires that strip slopes of vegetation, and human activity such as excavation or construction on unstable slopes.

In India, roughly 12-15% of the country's land area is considered landslide-prone, concentrated mainly in the Himalayan and sub-Himalayan states, the Northeastern hill ranges, and the Western Ghats. The steep terrain, high-intensity monsoon rainfall, and ongoing seismic activity in these regions combine to make landslides a recurring, often monsoon-season, hazard.

2. Warning Signs to Watch For

Landslides do not always give clear warning, but in many cases the ground shows signs of instability in the hours, days, or weeks beforehand. Learn to recognize these signs, especially if you live in or are traveling through a known landslide-prone area:

- New cracks or unusual bulges appearing in the ground, in street pavement, or in your home's floors and walls.
- Doors or windows that stick or jam for the first time, and visible gaps appearing around door and window frames.
- Soil visibly pulling away from foundations, or small landslides or debris flows occurring upslope.
- Fences, retaining walls, utility poles, or trees tilting, moving, or leaning in the same direction.
- Underground utility lines breaking, or new water breakouts from the ground in places that weren't previously wet.
- A sudden increase or decrease in water flow in a nearby stream or creek, especially if the water changes from clear to muddy.
- Unusual sounds — a faint rumbling that grows louder, trees cracking, or boulders knocking together — which can indicate moving debris.

3. Before a Landslide — Preparedness

Do's

- Find out whether your area has a history of landslides, or is officially mapped as landslide-prone, by checking with local disaster management authorities.
- Plan an evacuation route in advance, ideally more than one, and agree on a family meeting point away from slopes and drainage channels.
- Keep an emergency kit ready with a torch, first-aid supplies, drinking water, a battery-powered or hand-crank radio, important documents, and any regular medication.

- Keep drains, culverts, and channels near your home clear of litter, leaves, and debris so water can flow away freely during heavy rain.
- Plant trees and deep-rooted vegetation on slopes near your property — root systems help bind soil and reduce the chance of slope failure.
- During heavy or prolonged rainfall, listen to official weather bulletins and follow evacuation advisories immediately if issued.
- If you must travel through hilly or mountainous terrain, check weather and road advisories beforehand and avoid travel during or immediately after intense rainfall.

Don'ts

- Don't build homes or other structures directly on steep slopes, at the base of steep slopes, or within natural drainage channels and debris-flow paths.
- Don't ignore small, recurring slope movements or minor cracks — they often precede a larger failure.
- Don't rely solely on hearing a rumbling sound or seeing sliding debris as your only warning — a landslide can occur suddenly with no perceptible lead time.
- Don't block or divert natural drainage paths near your property without proper engineering guidance, as this can concentrate water and destabilize slopes.

4. During a Landslide — Immediate Response

Do's

- Move away from the path of the landslide or debris flow as quickly as possible — moving perpendicular to (across) the direction of movement, not directly away from it if that means running downhill in its path.
- Get to higher ground, away from the slope and away from stream channels, drainage paths, and areas that channel debris flows.
- If escape is not possible and you are indoors, move to the highest available floor and away from the side of the building facing the slope.
- If you're driving, stay alert for collapsed pavement, mud, fallen rocks, and other signs of debris flow on the road, and be ready to stop or turn back safely.
- If you are trapped and cannot move, protect your head and airway, cover your mouth and nose with cloth to reduce dust inhalation, and try to signal your location — a tapping sound on a hard surface or a whistle can carry further and cost less energy than shouting.
- Call your national emergency number (112 in India) as soon as it is safe to do so, and give your exact location.

Don'ts

- Don't return to or linger in a building or area that shows active signs of ground movement.
- Don't attempt to outrun a fast-moving debris flow downhill in the direction it is moving — move across its path instead.
- Don't cross a bridge if you can see or hear a debris flow approaching it.
- Don't waste time collecting belongings once evacuation is necessary — leave immediately.

5. After a Landslide — Staying Safe

Do's

- Stay away from the slide area — additional slope failure or flash flooding often follows the initial event.
- Check yourself and others nearby for injuries, and provide first aid or call for medical help as needed.
- Report broken utility lines (electrical, gas, water, sewage) to the relevant authority, and avoid the immediate area.
- Check the foundation of your home and surrounding land for cracks or other damage before re-entering — if in doubt, have it inspected by a qualified professional first.
- Watch for flooding, which can occur after a landslide, especially if the slide blocked a stream, culvert, or drainage channel.
- Replant damaged ground as soon as possible once it's safe to do so — bare slopes are significantly more vulnerable to erosion and further slides in subsequent rainfall.

Don'ts

- Don't re-enter your home or the affected area until local authorities have confirmed it's safe to do so.
- Don't assume the danger has passed immediately after a slide — secondary slides and flash flooding are common follow-on hazards.
- Don't disturb or attempt to move debris that may be unstable or that could be covering a person or utility line.

6. Special Guidance for Landslide-Prone Regions in India

The National Disaster Management Authority (NDMA) maintains Landslide Hazard Zonation (LHZ) maps that identify areas at repeated risk. Residents of hilly regions in Uttarakhand, Himachal Pradesh, Jammu & Kashmir, the Northeastern states, and the Western Ghats states (Kerala, Karnataka, Tamil Nadu, Goa, Maharashtra) should treat the monsoon season as a period of heightened landslide risk. Local disaster management authorities issue landslide warnings and advisories through the India Meteorological Department and state disaster management channels — following these promptly, rather than waiting for visible signs, is the most reliable way to stay safe.

In an active emergency, call 112 (India's national emergency number) immediately and give your exact location. This guide is general safety information and does not replace instructions from local emergency authorities.